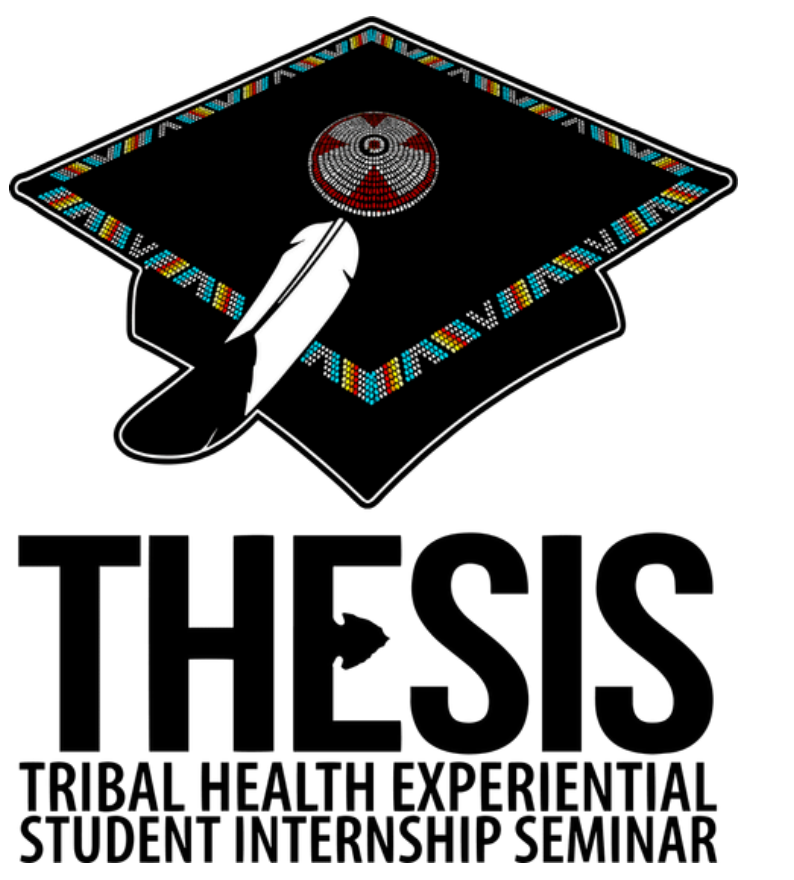




Influence of Maternal Practices on Sudden Infant Death Syndrome in Native Americans



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Introduction

Sudden Infant Death Syndrome (SIDS) is the sudden and unexpected death of an infant under one year of age during sleep.¹ Native American infants, especially those in Northern regions, experience a higher incidence of SIDS compared to their white counterparts.² This research aims to investigate the contributing factors, including maternal practices such as smoking habits, the sleep position of the infant, and breastfeeding practices, to understand and address the elevated risk of SIDS in Native American communities.

Objectives

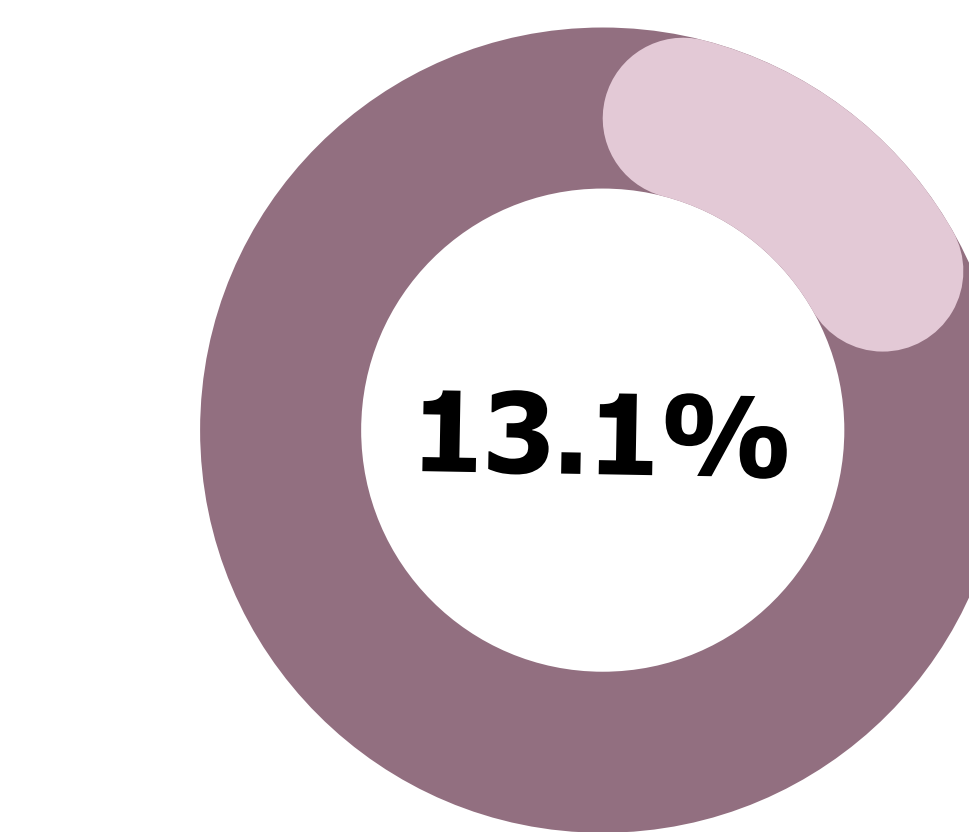
- Explain the elevated risk of Sudden Infant Death Syndrome (SIDS) among Native American populations.
- Provide key practices to adopt and behaviors to avoid to reduce the risk of Sudden Infant Death Syndrome (SIDS).
- Improve the overall maternal health of Native Americans by advocating for increased funding and greater involvement from public health professionals.

Methodology

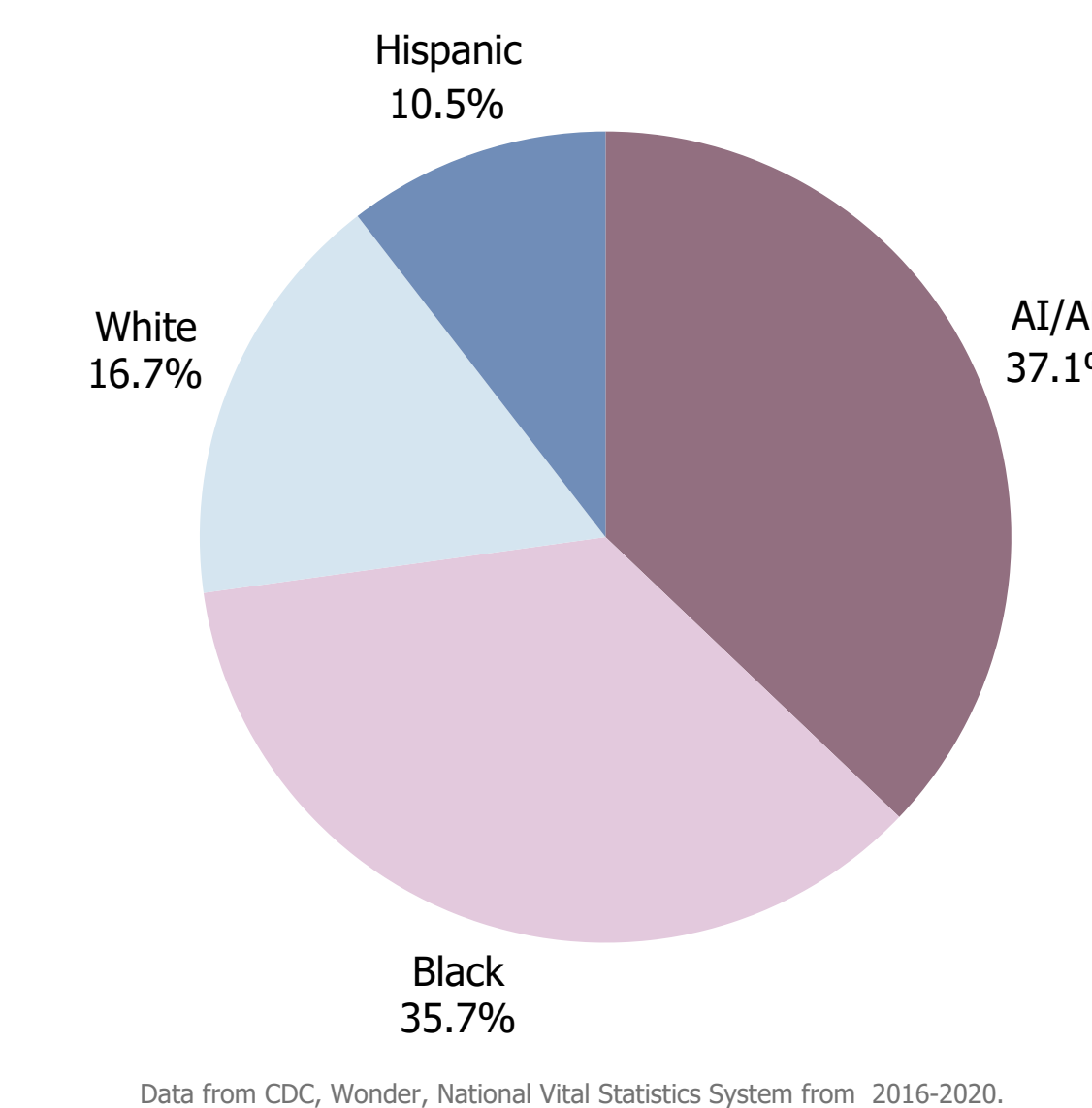
Literature Review: Performed an extensive review on the influence of maternal practices on the prevalence of Sudden Infant Death Syndrome.

Leveraged datasets to examine the number of cases among Native Americans.

Analysis

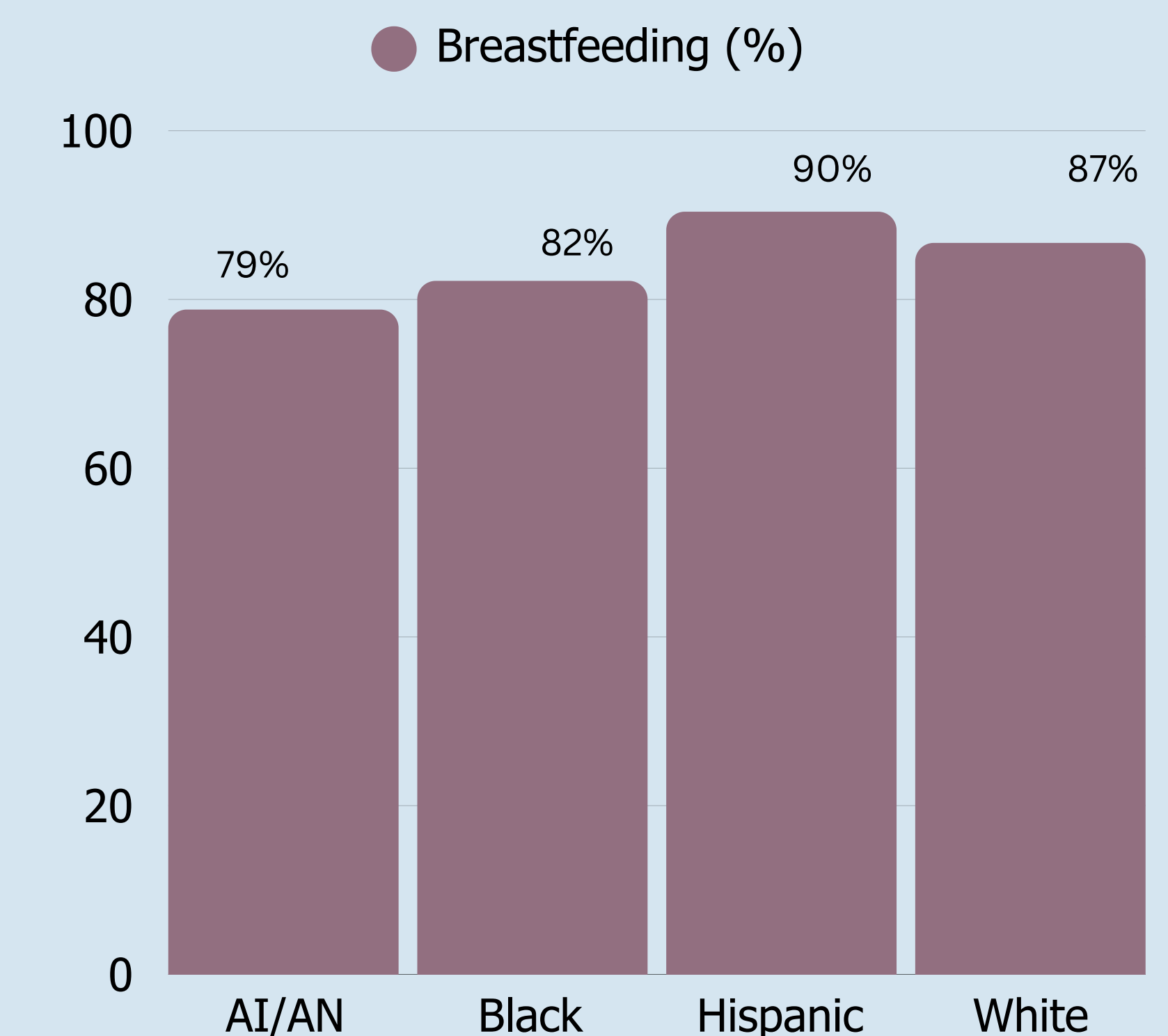


13.1% of Native Americans participate in safe sleep practices in Oklahoma.



Data from CDC, Wonder, National Vital Statistics System from 2016-2020.

Figure 1. Breastfeeding Rates in Oklahoma, 2021



Data from CDC, Pregnancy Risk Assessment Monitoring System, Oklahoma 2021

Results/Findings

Sleeping Position

The study found that Native American infants in North Dakota have a significantly higher relative risk for Sudden Infant Death Syndrome (SIDS) compared to white infants, with a relative risk of 3.2. Despite this elevated risk, only 46.9% of Native American infants slept in the prone position, compared to 74.4% of white infants. This suggests that while prone sleeping is a known risk factor for SIDS, other factors such as the use of certain mattresses, and overdressing may also contribute to the higher SIDS rates among Native American infants.³

Breastfeeding Practices

Figure 1 indicates that Native Americans when compared to other races have the least amount of mothers practicing breastfeeding. Breastfeeding lowers the risk of Sudden Infant Death Syndrome (SIDS) by providing essential antibodies that protect against infections and promoting frequent arousal from sleep, which helps prevent airway compromise. Studies have shown that infants who are formula-fed are twice as likely to die of SIDS compared to breastfed infants.² Additionally, the physical closeness during breastfeeding enhances maternal monitoring and response. Overall, breastfeeding offers comprehensive protective benefits that significantly reduce the risk of SIDS.²

Maternal Smoking

Northern American Indians and Alaska Natives (AI/AN) have alarmingly high maternal smoking rates, ranging from 44 to 67 percent, compared to only 13 percent among Southwestern Indians like the Navajo. This difference contributes to a higher SIDS incidence of 4.6 per 1,000 live births in the Northern region versus 1.4 per 1,000 in the Southwest. Maternal smoking during pregnancy increases the risk for SIDS by causing chronic fetal hypoxia and impairing fetal central nervous system development.⁴

Safe Sleep Practices

Compared to other races, Native Americans have the lowest percentage of mothers participating in safe sleep practices in Oklahoma. Examples of safe sleep practices that reduce the risk of SIDS include placing the child on their back to sleep, the baby with its own space to sleep, and having a space free of blankets and other necessities. Furthermore, 21% of the Native American infants slept with a pillow, compared to less than 2% of white infants.³ Co-sleeping on sofas, couches, or recliners is highly dangerous and should always be avoided. These environments pose a high risk of accidental suffocation or asphyxiation.² Lastly, American Indian/Alaskan Native groups may be aware of Infant Safe Sleep (ISS) guidelines but prefer to follow infant sleep practices that are deemed "unsafe" due to cultural preferences. This makes it challenging for providers to engage in conversations around shifting behavior.

Conclusion

- This study is relevant as it addresses the high incidence of Sudden Infant Death Syndrome (SIDS) among Northern American Indians and Alaska Natives, emphasizing the need for targeted interventions. Increasing breastfeeding rates can be achieved through culturally sensitive education, peer counseling, and community support initiatives. Reducing maternal smoking requires targeted, accessible smoking cessation programs involving community leaders and healthcare providers. Promoting safe sleep practices can be enhanced through public health campaigns that educate parents on safe sleep environments, and delivered with trusted community channels. Addressing these areas can help reduce SIDS rates among Northern American Indian and Alaska Native infants.
- Research on Sudden Infant Death Syndrome (SIDS) in Native American communities is hindered by limited access to comprehensive health data and small sample sizes due to the diverse and often small population of tribes. Cultural variations in infant care practices and the impact of historical trauma further complicate understanding and addressing SIDS risk factors effectively.
- Future research on Sudden Infant Death Syndrome (SIDS) among Native Americans should prioritize genetic predispositions, cultural beliefs through qualitative interviews, and tribe-specific studies to understand prevalence and contributing factors. These efforts aim to improve prevention and healthcare tailored to Native American communities.

References and Acknowledgement



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